

Section :- B

Q:2(i)

In 1803, John Dalton the British chemist presented a scientific theory on atoms. He named the theory "Dalton Atomic theory". His main postulates are as follows:-

- * All elements ~~particles~~ are made up of tiny indivisible particles called atoms.
- * Atoms of a ~~particular~~ particular element are identical. They have same mass and volume.
- * During chemical reactions, atoms combine or rearrange or separate in simple ratios.
- * Atoms can neither be created nor be destroyed.

Importance :-

It is considered the foundation of modern chemistry because it introduced the concept of the atom as the building block or monomer of an atom.

1. Monday

Q.2(ii)
Q.2(ii)

Atomic

Number

Mass

number

Definition

1) Total number of protons in the nucleus of an atom is called atomic number.

Total numbers of protons and neutrons present in nucleus is called mass number.

Symbol

* It's symbol is Z

It's symbol is 'A'.

Example
Determines

* $^{23}_{11}\text{Na}$: 11 is the atomic no.

$^{23}_{11}\text{Na}$: $23 - 11$
= 20

Q.2(iii)

Isotopes:-

Isotopes are atoms of an element whose nuclei have the same atomic number but different mass numbers. This is because atoms of an element can differ in the number of neutrons.

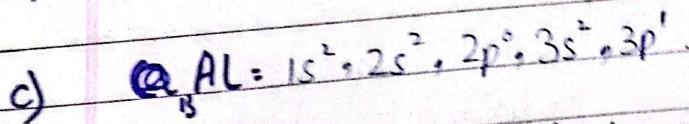
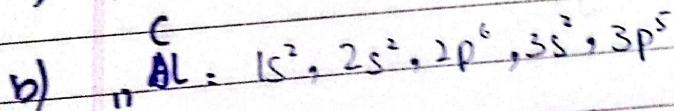
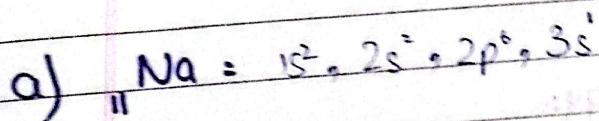
Isotopes of Hydrogen

Isotopes Hydrogen-1 Hydrogen-2 Hydrogen-3
 (Protium) (Deuterium) (Tritium)

Symbol	${}^1_1\text{H}$	${}^2_1\text{H}$	${}^3_1\text{H}$
N.O of protons	1	1	1
N.O of electrons	1	1	1
N.O of neutrons	0	1	2

Q.2 (iv)(a)

A:2 (iv)(or)



Q.2 (v)(or)

A:2 (v)(or)

(P.T.O)

Group

Period

Definition:

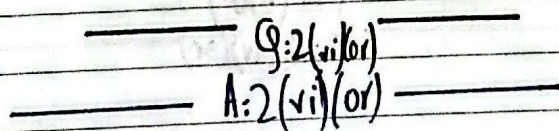
* Vertical rows of a periodic table are called groups.

The horizontal rows of a periodic table are called periods.

* There are 1-18 rows from left to right.

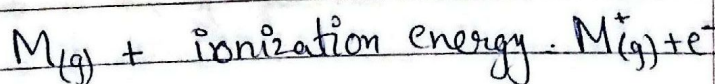
There are only 7 rows from top to bottom.

* Group number indicates the number of valence electron in an element. The properties of elements within a period change gradually.



✓ Ionization Energy:-

Ionization energy is a energy defined as the minimum amount of energy required to remove the outermost electron from an isolated gaseous atom.



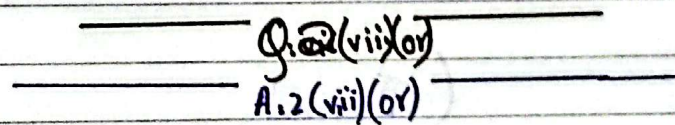
trends in period :-

Period :-

As you move from left to right in a period the ionization energy ~~and~~ increases.

trends in groups :-

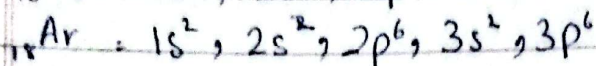
Ionization energy from top to bottom in a group:



Noble gases :-

There are eight groups of normal elements (1, 2, 3, 4, 5, 6, 7, 8) in the periodic table. Group 18 consists of noble gases or zero group elements because they are all very stable and chemically inert under normal condition. They exist in atmospheric form in the atmosphere. They have general electronic configuration: ns^2, np^6 (8 electron in valence shell) except $He (1s^2)$. These noble gases ~~are~~ have completely filled valence shell (s and p subshell). Their octet is complete, so they do not participate in ordinary chemical reaction & are called inert gases. They have 8

electrons in their valence shell except H, which has 2 electrons in its valence shell.



Q: 2(viii)

A: 2(viii)

Metals

Non-metals

(Thermal
conductivity)

* Metal generally have high thermal conductivity which means they can conduct heat easily.

* Non-metal tends to have poor thermal conductivity, making them less efficient at conductivity conducting heat.

Electrical conductivity

* Metals are good conductors of electricity because they have free electrons that can move freely.

Non-metals are poor conductors of electricity because they lack free electrons.

Adaptability

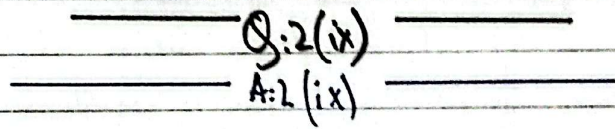
* Metals are ductile and malleable.

Non-metals are not ductile and malleable.

Melting points and boiling points

* Metals ^{have} are high melting and boiling points.

Non-metals have lower melting and boiling points.



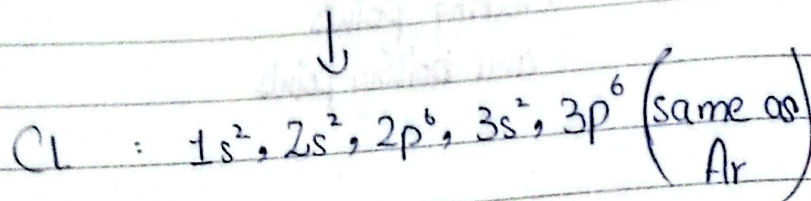
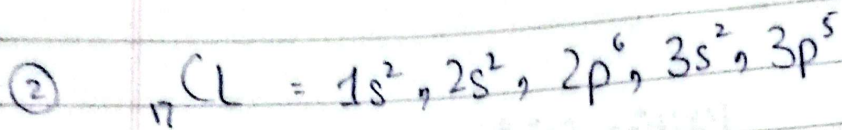
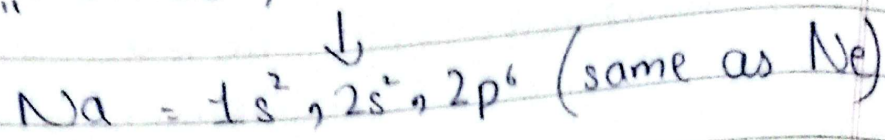
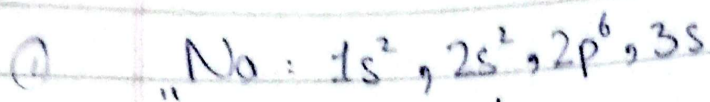
Octet rule :-

Atoms of the major group elements tend to obey octet rule in order to participate in chemical bonding in the form of eight electron per atoms in the resulting molecule. This rule only appears applies to the major group elements. The chemical behaviour of the main group elements can be predicted with the help of the octet rule. This is because the rule only involves s and p electron.

Day: _____

Date: _____

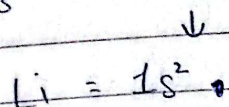
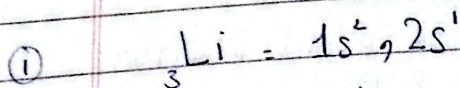
For example :-



Duplet rule:

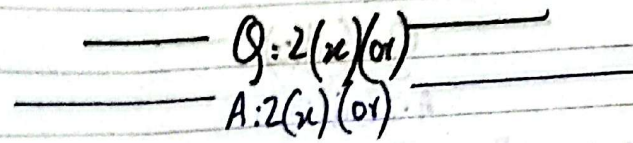
The tendency of atoms to acquire two electronic configuration in their outermost shell during bond formation is called duplet rule. They attain electronic configuration like helium.

For Example :-



Day: _____

Date: _____



Polar Covalent
Bond

Non-polar
covalent bond

Definition

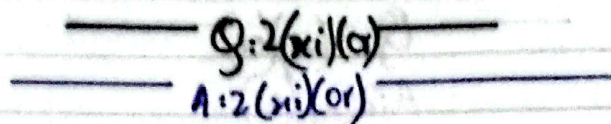
- | | |
|--|--|
| <p>* Bond formed b/w two different atoms with different electronegativities; electrons are shared unequally.</p> | <p>Bonds formed b/w two same atoms with same electronegativities where electrons are shared equally.</p> |
|--|--|

Charge Distribution

- | | |
|---|--|
| <p>* Same One end is δ^+, other end is δ^-</p> | <p>On Symmetrical, no partial charge.</p> |
|---|--|

Example

- | | |
|---|---|
| <p>* H-H</p> <p>* O=O</p> <p>* Cl=Cl</p> <p>* N≡N</p> | <p>* Cl-H</p> <p>* H-O-H</p> <p>* HCN</p> <p>* NH₃</p> |
|---|---|



Hydrogen Bonding:-

The interaction of a highly electron deficient hydrogen and lone pair on a nearly highly electronegative atom such as N, O, or F. is called hydrogen bonding.

hydrogen bonding in water:-

In a water molecule (H_2O), oxygen is highly electronegative and pulls the shared electrons toward itself, leaving the hydrogen atoms with a partial positive charge (δ^+) and oxygen with a partial negative charge (δ^-).

The δ^+ hydrogen of one water molecule is then weakly attracted to lone pair ~~on~~ on the oxygen of another water molecule, forming a hydrogen bond. This is responsible for water's unusually high boiling point, surface tension and its role as the universal solvent.

Section - C

Q:3(i)(or)
A:3(i)(or)

* Rutherford Experiment in 1911 :-

Rutherford bombarded a very thin layer of gold foil about 0.0004 cm thickness with α -particles. He used α -particles obtained from the disintegration of polonium. α -particles are helium nuclei that are double ~~part~~ positively charged (He^{++}). Most of these particles were slightly deflected. Around one in 1 million was deflected through an angle greater than 90° from their straight path.

✓ Rutherford made following conclusions:-

- i) Since majority of the α -particles passed through the foil undeflected, most of the space occupied by an atom must be empty.
- ii) The deflection of a few α -particles through angles greater than 90° shows that these particles are deflected by electrostatic repulsion between the positively charged α -particles

charged part of atom.

- 3) Massive α -particles are not deflected by electrons.

* Defects in Rutherford model:-

- 1) Classical physics suggests that electron being charged particles will emit energy continuously while revolving ~~and~~ around the nucleus. Thus, the orbit of the revolving electron becomes smaller and smaller until it would fall into the nucleus. This would collapse the atomic structure.

- 2) If revolving electron emits energy continuously it should form a continuous spectrum.

Q:3(ii)(or)

A:3(ii)(or)

Isotopes:-

Isotopes are atoms of an element whose nuclei have the same atomic number but different mass number. This is because atoms of an element

can differ in the number of neutrons.

✓ Isotopes of carbon :-

Isotope	Carbon-12	Carbon-13	Carbon-14
Symbol	${}^1_6\text{C}$	${}^{13}_6\text{C}$	${}^{14}_6\text{C}$
N.o of protons	6	6	6
N.o of electrons	6	6	6
N.o of neutrons	6	7	8
Natural abundance	98.9%	1.1%	0.0009%

✓ Isotope of Hydrogen

Isotopes	Hydrogen-1 (protium)	Hydrogen-2 (deuterium)	Hydrogen-3 (tritium)
Symbol	${}^1_1\text{H}$	${}^2_1\text{H}$	${}^3_1\text{H}$
N.o of protons	1	1	1
N.o of electrons	1	1	1

Day: _____

Date: _____

N.o of
neutrons

0

1

2

N.A

99.99%

0.0015%

Rare

Isotopes of chlorine :-

Isotopes

Chlorine-35

Chlorine-37

N.o of
protons

17

17

Symbol

 $^{35}_{17}\text{Cl}$ $^{37}_{17}\text{Cl}$ N.o of
electron

17

17

N.A

75.77%

24.23%

N.o of
neutrons

18

20

Q: 3(iii)
A: 3(iii)

= Atomic size :-

The average distance
b/w the nucleus and the
outermost electron shell.

★ Across ^ato period :- It decreases,
because the nuclear charge
increases while electrons are
added to the same valence
shell, pulling them closer to the
nucleus.

★ Down a group :- It decreases,
because new electron
shells are added with each
successive element.

= Ionization Energy :-

The minimum
energy required to remove the
outermost electron from
an isolated gaseous atom.

Across a period :- Increases, because
the nuclear charge increases
and atomic size decreases, ~~holding~~

valence electrons more tightly.

✓ Down a group :-

Decreases, because atomic size and shielding effect increase so the valence electron is held more loosely.

* Electron affinity :-

The amount of energy that is released when an electron is added to the valence shell of an isolated gaseous atom.

✓ Across a period :-

Increases due to higher nuclear charge and smaller atomic radius.

✓ Down a group :-

Decreases, due to increased shielding effect and atomic radius.

✓ Electronegativity :-

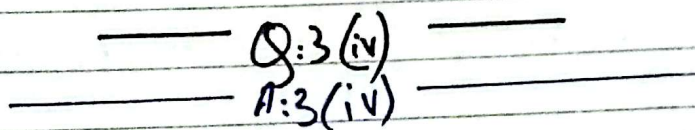
The ability of an atom to attract shared electrons toward itself in a chemical bond.

Trend in period:-

F has the highest value (4.0)

Trend in group:-

Cs has one of the lowest value (0.7)



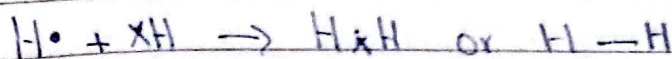
Covalent Bond:-

A chemical bond formed by the mutual sharing of electron between two non-metal atoms is called a covalent bond.

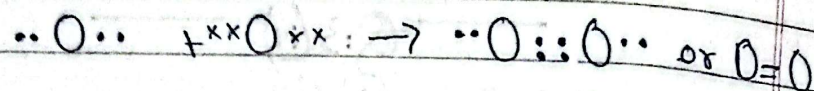
A covalent bond is represented by a single line (single bond), double line (double bond), or triple line (triple bond) between the bonded atoms.

1) **Single covalent bonds - H_2 :-**

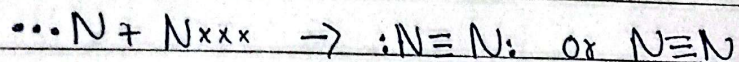
Each hydrogen atom has 1 valence electron. To attain the helium duplet, two H atoms share one pair of electrons forming a single covalent bond.



- 2) Double covalent bond - O_2 :-
 Oxygen (Group 16) has 6 valence electrons and need 2 more to complete its octet. Two oxygen atoms share two pairs of electrons, forming a double covalent bond :-



- 3) Triple covalent bond - N_2 :-
 Nitrogen (Group 5) has 5 valence electrons and need 3 more electrons to complete its octet. Two Nitrogen atoms share three pairs of electrons, forming triple covalent bond.



• It is a strong bond.

MCQs:

- 1) C (neutron)
- 2) B (protons)
- 3) C (20)
- 4)  B (6)
- 5) C (Alkali metals)
- 6) C (7)
- 7) B (Fluorine)
- 8) B (Decreases)
- 9) B (Ionic bond)
- 10) C (Three)
- 11) C (NH_4^+)
- 12) B ($\text{N}, \text{O}, \text{F}$)